Differentiation

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Implicit differentiation**

Find $\frac{dy}{dx}$ given $x^3 + y^3 = 6xy$, and evaluate it at the point $(3, 3)$.

Differentiate: $3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$. Rearranging: $\frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2$, so $\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$. At $(3, 3)$: $\frac{2(3) - 9}{9 - 2(3)} = \frac{-3}{3} = -1$.

2 **[Foundational]** • **Target: Implicit differentiation**

Find $\frac{dy}{dx}$ given $\sin(xy) = x$, using implicit differentiation.

Differentiate both sides: $\cos(xy)\left(y + x\frac{dy}{dx}\right) = 1$. Expanding: $y\cos(xy) + x\cos(xy)\frac{dy}{dx} = 1$. Rearranging: $\frac{dy}{dx} = \frac{1 - y\cos(xy)}{x\cos(xy)}$.

3 **[Foundational]** • **Target: Parametric differentiation**

A curve is defined parametrically by $x = t^2$, $y = 2t^3$. Find $\frac{dy}{dx}$ in terms of t.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 6t^2$. $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{6t^2}{2t} = 3t$ (for $t \neq 0$).

4 **[Foundational]** • **Target: Parametric differentiation**

A curve is defined by $x = 3\cos t$, $y = 3\sin t$. Find $\frac{dy}{dx}$ in terms of t, and find the gradient at $t = \frac{\pi}{4}$.

$\frac{dx}{dt} = -3\sin t$, $\frac{dy}{dt} = 3\cos t$. $\frac{dy}{dx} = \frac{3\cos t}{-3\sin t} = -\cot t$. At $t = \frac{\pi}{4}$: $-\cot\left(\frac{\pi}{4}\right) = -1$, since $\cos\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right)$.

5 **[Foundational]** • **Target: Related rates**

A spherical balloon is inflated so its volume increases at a constant rate of 100 cm^3/s. Find the rate of increase of the radius when $r = 5$ cm, using $V = \frac{4}{3}\pi r^3$, to three decimal places.

$\frac{dV}{dr} = 4\pi r^2$. At $r = 5$: $\frac{dV}{dr} = 4\pi(25) = 100\pi$. $\frac{dr}{dt} = \frac{dV/dt}{dV/dr} = \frac{100}{100\pi} = \frac{1}{\pi} \approx 0.318$ cm/s.

6 **[Foundational]** • **Target: Related rates**

The area of a circular ripple increases at 8π cm^2/s. Find the rate of change of the radius when $r = 4$ cm, using $A = \pi r^2$.

$\frac{dA}{dr} = 2\pi r$. At $r = 4$: $\frac{dA}{dr} = 8\pi$. $\frac{dr}{dt} = \frac{dA/dt}{dA/dr} = \frac{8\pi}{8\pi} = 1$ cm/s.

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7 [Foundational] • Target: Related rates

A cone-shaped tank (vertex down) has height 10 cm and top radius 4 cm, so at water depth h the radius is $r = \frac{2h}{5}$. Water is poured in at 12 cm^3/s. Find the rate of rise of the water level when $h = 5$ cm, using $V = \frac{1}{3}\pi r^2 h$.

Substitute $r = \frac{2h}{5}$: $V = \frac{1}{3}\pi\left(\frac{2h}{5}\right)^2 h = \frac{4\pi}{75}h^3$. $\frac{dV}{dh} = \frac{12\pi}{75}h^2 = \frac{4\pi}{25}h^2$. At $h = 5$: $\frac{dV}{dh} = \frac{4\pi}{25}(25) = 4\pi$. $\frac{dh}{dt} = \frac{dV/dt}{dV/dh} = \frac{12}{4\pi} = \frac{3}{\pi} \approx 0.955$ cm/s.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Related rates

The radius of a circular oil slick increases at a constant rate of 0.5 m/min. Find the rate of increase of its area when $r = 20$ m.

$A = \pi r^2$, so $\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$. At $r = 20$: $\frac{dA}{dt} = 2\pi(20)(0.5) = 20\pi \approx 62.83$ m^2/min.

9 [Intermediate] • Target: Applying the rules: tangents and second derivatives

Find the equation of the tangent to $y = xe^{-x}$ at $x = 1$.

By the product rule: $\frac{dy}{dx} = e^{-x} + x(-e^{-x}) = e^{-x}(1 - x)$. At $x = 1$: $\frac{dy}{dx} = e^{-1}(0) = 0$. Since $y(1) = e^{-1}$, the tangent is horizontal: $y = \frac{1}{e}$.

10 [Intermediate] • Target: Applying the rules: tangents and second derivatives

Find $\frac{d^2y}{dx^2}$ for $y = \ln(x^2 + 1)$, and evaluate it at $x = 1$.

$\frac{dy}{dx} = \frac{2x}{x^2+1}$. By the quotient rule: $\frac{d^2y}{dx^2} = \frac{2(x^2+1) - 2x(2x)}{(x^2+1)^2} = \frac{2 - 2x^2}{(x^2+1)^2}$. At $x = 1$: $\frac{2-2}{4} = 0$.

11 [Intermediate] • Target: A well-designed word problem: a ladder sliding down a wall

This question has three parts. A ladder 5 m long leans against a vertical wall, its base on horizontal ground. The base slides away from the wall at a constant rate of 0.4 m/s. (a) Let x be the distance of the base from the wall and y the height of the top of the ladder. Write the relationship between x and y, and differentiate implicitly with respect to time t to relate $\frac{dx}{dt}$ and $\frac{dy}{dt}$. (b) Find the rate at which the top of the ladder is sliding down the wall when the base is 3 m from the wall. (c) Using $x = 5\cos\theta$, where θ is the angle between the ladder and the ground, find the rate at which θ is changing at that instant.

(a) $x^2 + y^2 = 25$. Differentiating with respect to t: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$, so $\frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}$. (b) When $x = 3$, $y = \sqrt{25 - 9} = 4$. $\frac{dy}{dt} = -\frac{3}{4}(0.4) = -0.3$ m/s, so the top slides down at 0.3 m/s. (c) Differentiating $x = 5\cos\theta$: $\frac{dx}{dt} = -5\sin\theta\frac{d\theta}{dt}$. At this instant $\sin\theta = \frac{y}{5} = \frac{4}{5} = 0.8$, so $0.4 = -5(0.8)\frac{d\theta}{dt} = -4\frac{d\theta}{dt}$, giving $\frac{d\theta}{dt} = -0.1$ rad/s (the angle is decreasing).

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12 **[Intermediate]** • **Target: The quotient rule**

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ of $y = x^{\sin x}$ for $x > 0$ using logarithmic differentiation.

Take natural logarithms of both sides: $\ln y = \ln(x^{\sin x}) = \sin x \cdot \ln x$. Differentiate implicitly with respect to x using the Product Rule on the right: $\frac{1}{y}\frac{dy}{dx} = (\cos x)(\ln x) + (\sin x)\left(\frac{1}{x}\right)$. Multiply both sides by $y = x^{\sin x}$: $\frac{dy}{dx} = x^{\sin x}\left(\cos x \ln x + \frac{\sin x}{x}\right)$.

13 **[Intermediate]** • **Target: Implicit differentiation**

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

Express $y = (x^4 + 3x^2 + 7)^{1/2}$. By the Chain Rule: $y' = \frac{1}{2}(x^4 + 3x^2 + 7)^{-1/2} \cdot \frac{d}{dx}(x^4 + 3x^2 + 7) = \frac{4x^3 + 6x}{2\sqrt{x^4 + 3x^2 + 7}} = \frac{2x^3 + 3x}{\sqrt{x^4 + 3x^2 + 7}}$. Evaluate at $x = 1$: $y'(1) = \frac{2(1)^3 + 3(1)}{\sqrt{1^4 + 3(1)^2 + 7}} = \frac{5}{\sqrt{11}} = \mathbf{\frac{5\sqrt{11}}{11}}$.

14 **[Intermediate]** • **Target: Parametric differentiation**

Level 3 · Advanced A particle moves along a curve defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the value of $\frac{d^2y}{dx^2}$ at $t = 2$ and determine whether the curve is concave up or concave down at this point.

First find the components: $\frac{dx}{dt} = 2t$ and $\frac{dy}{dt} = 3t^2 - 3$. The first derivative is $\frac{dy}{dx} = \frac{3t^2 - 3}{2t} = \frac{3}{2}t - \frac{3}{2t}$. Differentiating with respect to t: $\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{3}{2} + \frac{3}{2t^2}$. Then $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{dx/dt} = \frac{\frac{3}{2} + \frac{3}{2t^2}}{2t}$. At $t = 2$: numerator is $\frac{3}{2} + \frac{3}{8} = \frac{15}{8}$; denominator is $2(2) = 4$. Thus.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Related rates** • **Cross-Lesson: 5.01 Integration**

Level 3 · AdvancedEvaluate the indeterminate limit using repeated applications of L'Hôpital's Rule with complete hypothesis verification: $\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.

Step 1 · Initial Verification: As $x \to 0$, numerator $0 - 0 = 0$ and denominator $0^3 = 0$. Form $\frac{0}{0}$. Differentiating numerator and denominator: $\lim_{x \to 0} \frac{1 - \frac{1}{1+x^2}}{3x^2} = \lim_{x \to 0} \frac{\frac{(1+x^2)-1}{1+x^2}}{3x^2} = \lim_{x \to 0} \frac{x^2}{3x^2(1+x^2)}$. Step 2 · Algebraic Simplification: For $xeq0$, cancel the common factor x^2: $= \lim_{x \to 0} \frac{1}{3(1+x^2)} = \frac{1}{3(1+0)} = \mathbf{\frac{1}{3}}$.

16 **[Challenging]** • **Target: Differentiating trigonometric, exponential and logarithmic composites** • **Cross-Lesson: 6.01 Sequences & Series**

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

Combine the logarithms on the left-hand side using the product rule: $\log_3[x(x + 6)] = 3$. Convert to exponential form: $x(x + 6) = 3^3 = 27 \Rightarrow x^2 + 6x - 27 = 0$. Factor the quadratic: $(x + 9)(x - 3) = 0 \Rightarrow x = -9$ or $x = 3$. Check the domain restrictions in the original equation:For $x = -9$: $\log_3(-9)$ is undefined, so $x = -9$ is an extraneous solution.For $x = 3$: $\log_3(9) + \log_3(3) = 2 + 1 = 3$. Valid ✓.Therefore, the unique solution is $x = 3$.

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17 [Challenging] • Target: Applying the rules: tangents and second derivatives • Cross-Lesson: 7.01 Vectors

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

(a) Find $\frac{dy}{dx}$: $\frac{dy}{dx} = 6x^2 - 18x + 12$ Set $\frac{dy}{dx} = 0$: $6(x^2 - 3x + 2) = 0 \Rightarrow 6(x-1)(x-2) = 0$, giving $x = 1$ and $x = 2$. When $x = 1$: $y = 2(1) - 9(1) + 12(1) + 1 = 6$. Point is $(1, 6)$. When $x = 2$: $y = 2(8) - 9(4) + 12(2) + 1 = 16 - 36 + 24 + 1 = 5$. Point is $(2, 5)$. (b) Second derivative: $\frac{d^2y}{dx^2} = 12x - 18$ At $x = 1$: $\frac{d^2y}{dx^2} = 12(1) - 18 = -60$, so $(2, 5)$ is a local minimum.

18 [Challenging] • Target: A well-designed word problem: a ladder sliding down a wall • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

If 20% decays in 15 days, then 80% remains at $t = 15$: $0.80M_0 = M_0 e^{-15k} \Rightarrow e^{-15k} = 0.80$. Take the natural logarithm of both sides: $-15k = \ln(0.80) \Rightarrow k = -\frac{\ln(0.80)}{15} = \frac{\ln(1.25)}{15} \approx 0.01488$ day^{-1}. The half-life $t_{1/2}$ satisfies $e^{-kt_{1/2}} = 0.5$: $t_{1/2} = \frac{\ln 2}{k} = \frac{15 \ln 2}{-\ln(0.80)} \approx \frac{15(0.6931)}{0.2231} \approx 46.6$ days.

19 [Challenging] • Target: The chain rule • Cross-Lesson: 9.01 Complex Numbers

[Exponential, Logarithmic & Logistic Dynamics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: The product rule • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Level 3 · AdvancedEvaluate the limit analytically without using L'Hôpital's Rule: $\lim_{x \to 0} \frac{\sqrt{1+\tan x} - \sqrt{1+\sin x}}{x^3}$.

Step 1 · Rationalize the Numerator: Multiply by $\sqrt{1+\tan x} + \sqrt{1+\sin x}$: $\lim_{x \to 0} \frac{(1+\tan x) - (1+\sin x)}{x^3(\sqrt{1+\tan x} + \sqrt{1+\sin x})} = \lim_{x \to 0} \frac{\tan x - \sin x}{x^3(\sqrt{1+\tan x} + \sqrt{1+\sin x})}$. Step 2 · Factor Trigonometric Terms: $\tan x - \sin x = \frac{\sin x}{\cos x} - \sin x = \sin x\left(\frac{1 - \cos x}{\cos x}\right)$. Multiply numerator and denominator by $1 + \cos x$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The chain rule

Related rates: a 10-m ladder slides down a wall. When the base is 6 m from the wall and moving away at 2 m/s, the top is moving at:

a. $-\frac{3}{2}$ **m/s ✓**

b. $-\frac{3}{4}$ m/s

c. $\frac{3}{2}$ m/s

d. $\frac{4}{3}$ m/s

$x^2 + y^2 = 100$. At $x = 6$: $y = 8$. $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{6 \cdot 2}{8} = -\frac{3}{2}$ m/s.

22 [Foundational] • Target: The product rule

A ladder 10 m long leans against a wall; the base slides away at 1 m/s. How fast is the top sliding down when the base is 6 m from the wall?

a. $\frac{3}{4}$

b. $-\frac{3}{8}$

c. $-\frac{3}{4}$ ✓

d. $-\frac{3}{2}$

$x^2 + y^2 = 10^2$. At $x = 6$: $y = \sqrt{10^2 - 6^2} = 8$. Differentiating: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt} = -\frac{3}{4}$ m/s.

23 [Foundational] • Target: The quotient rule

Related rates: ladder 10 m against wall. Base slides at 1 m/s when base is 6 m. Rate of top:

a. $-3/4$ **m/s ✓**

b. $3/4$ m/s

c. -1 m/s

d. $-4/3$ m/s

$x^2 + y^2 = 100 \Rightarrow 2xy' + 2x\dot{x} = 0 \Rightarrow y' = -6/(8) \cdot 1 = -3/4$.

24 [Foundational] • Target: Implicit differentiation

Solve $x^2 - 2x + 17 = 0$. Find the product of the roots.

a. 16

b. 2

c. **17 ✓**

d. 18

Product of roots $= c/a = 17$ (roots are $1 \pm 4i$).

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25 [Foundational] • Target: Parametric differentiation

A ladder 13 m long leans against a wall. Base slides at 2 m/s. Rate of top falling when base is 5 m from wall:

a. $-\frac{5}{6}$ m/s ✓

b. $-\frac{5}{12}$ m/s

c. $-\frac{2}{5}$ m/s

d. $-\frac{1}{2}$ m/s

From $x^2 + y^2 = 169$, when $x = 5$ we have $y = 12$. Differentiating gives $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{5(2)}{12} = -\frac{5}{6}$ m/s.

26 [Foundational] • Target: Related rates

A 10-ft ladder leans against a wall; the base is 6 ft from the wall and slides away at $\frac{dx}{dt} = 1$ ft/s. Find $\frac{dy}{dt}$, the rate the top slides down (height $y = 8$ ft).

a. -1 ft/s

b. $-\frac{3}{4}$ ft/s ✓

c. $-\frac{1}{4}$ ft/s

d. $-\frac{1}{2}$ ft/s

Working through this calculus step: $x^2 + y^2 = 10^2 \Rightarrow 2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt} = -\frac{6}{8}(1) = -\frac{3}{4}$

27 [Foundational] • Target: Differentiating trigonometric, exponential and logarithmic composites

Implicit differentiation: $\frac{d^2y}{dx^2}$ at $(1, 1)$ for $x^2 + y^2 = 2$:

a. -1

b. -2 ✓

c. 0

d. 1

Differentiating $x^2 + y^2 = 2$ gives $y' = -\frac{x}{y}$ and $y'' = -\frac{y - xy'}{y^2} = -\frac{x^2 + y^2}{y^3}$. At $(1, 1)$, $y'' = -\frac{2}{1^3} = -2$.

28 [Foundational] • Target: Applying the rules: tangents and second derivatives

Related rates: water drains from a conical tank (height $H = 10$m, radius $R = 5$m) at 2 m³/min. When depth $h = 4$ m, $dh/dt =$

a. $-\frac{2}{4\pi}$ m/min

b. $-\frac{1}{2\pi}$ **m/min** ✓

c. $-\frac{2}{\pi}$ m/min

d. $-\frac{8}{\pi}$ m/min

$r = h/2$ (similar triangles). $V = \frac{\pi h^3}{12}$. $dV/dt = \frac{\pi h^2}{4} dh/dt = -2$. At $h = 4$: $dh/dt = -2/(4\pi) = -1/(2\pi)$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: A well-designed word problem: a ladder sliding down a wall**

Related rates: if $A = r^2\pi$, then $\frac{dA}{dt} =$

a. $2\pi r$

b. $2\pi r\frac{dr}{dt}$ ✓

c. $\pi r^2\frac{dr}{dt}$

d. $2\pi\frac{dr}{dt}$

Applying Applications of Derivatives principles, Chain rule. Therefore, the verified correct option is $2\pi r\frac{dr}{dt}$.

30 **[Intermediate]** • **Target: The chain rule**

Related rates: a balloon radius expands at 2 cm/s. Rate of volume change when $r = 5$ cm is:

a. 40π cm³/s

b. 100π cm³/s

c. 200π **cm³/s** ✓

d. 300π cm³/s

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2\frac{dr}{dt} = 4\pi(25)(2) = 200\pi$ cm³/s.

31 **[Intermediate]** • **Target: The product rule**

Logarithmic differentiation: if $y = \frac{x^2\sqrt{x+1}}{e^x}$, then $y' =$

a. $y\left(\frac{2}{x} + \frac{1}{2(x+1)} - 1\right)$ ✓

b. $y\left(\frac{2}{x} - \frac{1}{2(x+1)} - 1\right)$

c. $y\left(\frac{2}{x} + \frac{1}{2(x+1)} + 1\right)$

d. $y\left(2x + \frac{1}{2} - 1\right)$

$\ln y = 2\ln x + \frac{1}{2}\ln(x+1) - x$. Differentiate: $y'/y = 2/x + 1/(2(x+1)) - 1$.

32 **[Intermediate]** • **Target: The quotient rule**

Logarithmic differentiation: $y = x^x$. $y' =$

a. $x^x(1 + \ln x)$ ✓

b. $x \cdot x^{x-1}$

c. $x^x \ln x$

d. e^x

$\ln y = x\ln x \Rightarrow y'/y = 1 + \ln x \Rightarrow y' = x^x(1 + \ln x)$.

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33 [Intermediate] • Target: Implicit differentiation

Chain rule with two paths: $z = f(x, y)$, $x = r\cos\theta$, $y = r\sin\theta$. $\partial z/\partial r =$

a. $f_x\cos\theta + f_y\sin\theta$ ✓

b. f_r

c. $f_x r + f_y r$

d. $f_x\sin\theta - f_y\cos\theta$

Chain rule: $z_r = f_x x_r + f_y y_r = f_x\cos\theta + f_y\sin\theta$.

34 [Intermediate] • Target: Parametric differentiation

Implicit differentiation of $xy = 5$: $\frac{dy}{dx} =$

a. $\frac{5}{x}$

b. $\frac{y}{x}$

c. $-\frac{y}{x}$ ✓

d. $\frac{x}{y}$

$y + x\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -y/x$.

35 [Intermediate] • Target: Related rates

Using the Product Rule, $\frac{d}{dx}[(3x+1)(x^2-2)] =$

a. $3(x^2-2) + 2x(3x+1)$

b. $3x^2 - 6 + 6x^2 + 2x$

c. $9x^2 + 2x - 6$

d. **all are equivalent** ✓

All three expressions are equivalent: $9x^2 + 2x - 6$.

36 [Intermediate] • Target: Differentiating trigonometric, exponential and logarithmic composites

Related rates: sphere volume $V = (4/3)\pi r^3$. If $dr/dt = 2$ cm/s at $r = 3$ cm, $dV/dt =$

a. 72π **cm³/s** ✓

b. 36π cm³/s

c. 12π cm³/s

d. 24π cm³/s

$dV/dt = 4\pi r^2(dr/dt) = 4\pi \cdot 9 \cdot 2 = 72\pi$.

37 [Intermediate] • Target: Applying the rules: tangents and second derivatives

Implicit differentiation: $F(x, y) = 0$. Then $dy/dx =$

a. F_y/F_x

b. $-F_x/F_y$ ✓

c. F_x/F_y

d. $-F_y/F_x$

Implicit function theorem: $dy/dx = -F_x/F_y$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: A well-designed word problem: a ladder sliding down a wall** • **Cross-Lesson: 8.01 Algebra: Partial Fractions & Division**

A ladder 10 m long leans against a wall. If bottom slides out at 1 m/s, this is a:

a. linear rate

b. **related rates problem ✓**

c. optimisation

d. linearisation

Relationship between changing rates.

39 **[Challenging]** • **Target: The chain rule** • **Cross-Lesson: 9.01 Complex Numbers**

A geometric sequence has first term $a = 2$ and common ratio $r = 2$. Find the 5th term.

a. 64

b. -32

c. 34

d. **32 ✓**

$u_n = ar^{n-1} \Rightarrow u_5 = 2 \times (2)^4 = 32$.

40 **[Challenging]** • **Target: The product rule** • **Cross-Lesson: 1.01 Functions: Modulus & Inverse**

A line has equation $\mathbf{r} = (-2, 7, 12) + t(7, 4, 1)$. Find the x-coordinate of the point where $t = -5$.

a. -30

b. 7

c. -13

d. **-37 ✓**

At $t = -5$: point $= (-37, -13, 7)$. x-coordinate: -37.

41 **[Challenging]** • **Target: The quotient rule** • **Cross-Lesson: 10.01 Numerical Methods**

For the locus $|z - (3 + 4i)| = 4$, find the maximum value of $|z|$.

a. 1

b. 5

c. **9 ✓**

d. 4

$|3 + 4i| = 5$, so max $|z| = 5 + 4 = 9$ and min $|z| = 5 - 4 = 1$.

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42 [Challenging] • Target: Implicit differentiation • Cross-Lesson: 11.01 Differential Equations

A quantity decays according to $\frac{dy}{dt} = -ky$. If the quantity halves every 1 year, find k.

a. $\log(2) + 2$

b. $\log(2)$ ✓

c. $\log(3)$

d. $\log(2) + 1$

$e^{-kT} = \frac{1}{2} \Rightarrow k = \frac{\ln 2}{1}$.

43 [Challenging] • Target: Parametric differentiation • Cross-Lesson: 12.01 Poisson Distribution

A quantity decays according to $\frac{dy}{dt} = -ky$. If the quantity halves every 2 years, find k.

a. $2\log(2)$

b. $\frac{\log(3)}{2}$

c. $\frac{\log(2)}{2} + 1$

d. $\frac{\log(2)}{2}$ ✓

$e^{-kT} = \frac{1}{2} \Rightarrow k = \frac{\ln 2}{2}$.

44 [Challenging] • Target: Related rates • Cross-Lesson: 13.01 Continuous Random Variables

A line has equation $\mathbf{r} = (7, -7, 0) + t(2, -3, -4)$. Find the z-coordinate of the point where $t = 2$.

a. -13

b. 11

c. -12

d. **-8** ✓

At $t = 2$: point $= (11, -13, -8)$. z-coordinate: -8.

45 [Challenging] • Target: Differentiating trigonometric, exponential and logarithmic composites • Cross-Lesson: 14.01 Sampling & Estimation

A line has equation $\mathbf{r} = (2, -6, -2) + t(-4, -1, -4)$. Find the y-coordinate of the point where $t = -2$.

a. -5

b. 10

c. **-4** ✓

d. 6

At $t = -2$: point $= (10, -4, 6)$. y-coordinate: -4.

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